

Literature Review: Explainable Machine Learning for Irish Residential Energy Retrofit Prioritisation Using Building Energy Rating Data

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1 Introduction and Motivation

Improving residential energy efficiency is an important part of Ireland’s transition towards lower carbon emissions and more sustainable housing. Many existing homes differ in construction age, dwelling type, floor area, insulation quality, heating system, ventilation and energy performance. These differences mean that some homes require much more energy than others for space heating, water heating, ventilation and lighting.

In Ireland, the Building Energy Rating (BER) system provides a standardised method for assessing the energy performance of dwellings [1,4]. A BER rating allows homeowners, tenants, buyers and policymakers to compare homes on the basis of energy efficiency. Poor BER performance can indicate that a dwelling may benefit from energy retrofit measures, such as insulation upgrades, improved heating systems, better ventilation, or heat-pump installation.

The aim of this project is to develop an explainable machine-learning pipeline for Irish residential retrofit prioritisation using BER data. The project will investigate whether BER-related dwelling features can be used to predict which homes are likely to require retrofit priority, and which building characteristics are most strongly associated with poor energy performance.

2 Irish Retrofit and BER Context

A retrofit means upgrading an existing dwelling to improve its energy performance. Examples include attic insulation, wall insulation, window replacement, heating-control upgrades, ventilation improvements, solar photovoltaic installation, or replacing older heating systems with heat pumps. Since it is not practical to upgrade all inefficient homes immediately, retrofit prioritisation is a useful data-driven problem.

Ireland has national retrofit targets linked to improving residential energy efficiency. A commonly discussed policy target is upgrading the equivalent of 500,000 homes to BER B2 or cost-optimal level by 2030 [2,3], alongside large-scale heat-pump installation. This provides a clear policy motivation for identifying homes that are likely to have poor energy performance.

For this project, retrofit priority will be defined using BER rating information. A practical classification rule is:

- **Retrofit priority:** homes with BER worse than B2.
- **Not retrofit priority:** homes with BER B2 or better.

This definition is not intended to replace an official BER assessment or professional engineering

judgement. Instead, it provides a clear and policy-relevant modelling target for a computational project.

3 Data Source

The project will use publicly available or cleaned Irish Building Energy Rating data. BER datasets contain information about dwelling energy performance and construction features that affect energy efficiency. Relevant variables may include dwelling type, construction age band, floor area, wall characteristics, roof characteristics, heating system, fuel type, ventilation, primary energy use, CO₂ emissions and BER rating.

A cleaned Irish BER dataset published on Mendeley will be considered as the primary dataset. This is suitable because it supports reproducible analysis while avoiding reliance on restricted private transaction-level BER data. The dataset will first be inspected carefully to identify available columns, missing values, categorical variables, numerical variables and any features that could cause data leakage.

A key limitation of BER data is that it is based on standardised assessment assumptions rather than actual household energy consumption. Real energy use also depends on occupant behaviour, heating schedules and comfort preferences. Therefore, the project will be framed as BER-based energy-performance modelling, not as a direct prediction of household bills.

4 Planned Modelling Approach

The main modelling task will be binary classification. Let $Y_i = 1$ represent a dwelling requiring retrofit priority and $Y_i = 0$ represent a dwelling not requiring retrofit priority. Let x_i represent the feature vector for dwelling i .

A suitable baseline model is logistic regression:

$$P(Y_i = 1 \mid x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip})}}.$$

This model is appropriate because the target variable is binary and the model output can be interpreted as a probability of retrofit priority. Logistic regression also provides an interpretable mathematical baseline through its model coefficients.

The project will compare this baseline with tree-based machine-learning models such as decision trees, random forests and gradient boosting classifiers. These models can capture nonlinear relationships between building characteristics and retrofit-priority status. The purpose of the model comparison is to examine whether more flexible machine-learning models improve predictive performance while still allowing useful interpretation.

The models will be evaluated using standard classification metrics:

- accuracy,
- precision,
- recall,
- F1-score,
- ROC-AUC,
- confusion matrix.

Recall will be important because the model should identify as many poor-performing homes as possible. Precision will also be important because incorrectly prioritising too many homes may reduce practical usefulness.

5 Explainable Machine Learning

Explainability is an important part of this project because retrofit prioritisation is a decision-support problem. It is not enough to predict whether a home is a retrofit priority; the project should also explain which dwelling features are most associated with that prediction.

The project will use several explainability approaches. Logistic regression coefficients can show how individual features are associated with retrofit-priority probability. Tree-based feature importance and permutation importance can identify which variables most influence model performance. If technically stable, SHAP values may also be used to explain feature contributions in more detail.

The expected explainability analysis will answer questions such as:

- Which dwelling characteristics are most associated with poor BER performance?
- Are older buildings more likely to require retrofit priority?
- Do heating systems, dwelling type, floor area or construction features strongly influence retrofit-priority predictions?
- Can machine-learning models provide interpretable evidence for retrofit planning?

This makes the project more meaningful than a simple classification exercise, because it connects predictive modelling with practical insight.

6 Expected Project Workflow

The planned workflow is as follows:

1. Obtain and inspect the Irish BER dataset.
2. Select relevant dwelling and energy-performance variables.
3. Define the retrofit-priority target using the BER B2 threshold.
4. Clean the data and handle missing or inconsistent values.
5. Encode categorical variables and prepare numerical variables.
6. Perform exploratory data analysis on BER ratings, energy use, CO₂ emissions and dwelling characteristics.
7. Train logistic regression as an interpretable baseline.
8. Train tree-based models such as decision tree, random forest and gradient boosting.
9. Compare model performance using classification metrics.
10. Apply explainability methods to identify the most important predictors.
11. Summarise results in a GitHub README, final poster and presentation.

7 Expected Outcome

The expected outcome is a reproducible machine-learning pipeline that predicts whether Irish residential dwellings are likely to require retrofit priority based on BER-related features. The project should also identify the dwelling characteristics most associated with poor energy performance.

The final conclusion is expected to show whether BER data can support data-driven retrofit prioritisation and whether explainable machine-learning methods can provide useful insight into the factors influencing residential energy inefficiency. The project will be presented as a decision-support tool rather than a replacement for official BER assessment, engineering inspection or policy decision-making.

8 Conclusion

This literature review motivates an explainable machine-learning project for Irish residential energy retrofit prioritisation. The project is relevant because Ireland has strong residential retrofit targets and BER data provides structured information about dwelling energy performance. The proposed methodology combines statistical modelling, machine learning, model evaluation and explainability. This makes the project suitable for ACM 40960 because it includes mathematical modelling, computation, reproducible implementation and applied interpretation.

References

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